

Ministry of Higher Education

Soran University

Part 2 - Software Requirements Specification (SRS)

Second-hand Item Exchange System

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the **Second-hand Item Exchange System**. It defines what the system must do (functional requirements) and how well it must perform (non-functional requirements), so that the whole group, the supervisor, and later project parts (design, implementation, testing) share one clear reference.

1.2 Scope

The system is a web-based platform where students and community members can post used items, search for items, send buy or exchange requests, chat, rate each other, and where an administrator moderates the content. Online payment, delivery, and native mobile apps are outside the scope (see Section 8).

1.3 Definitions and Abbreviations

- **SRS** - Software Requirements Specification.
- **FR / NFR** - Functional Requirement / Non-Functional Requirement.
- **UR** - User Requirement (written from the user's point of view).
- **Listing** - An advertisement for a second-hand item posted by a user.
- **Request** - A buy or exchange request sent from one user to an item owner.
- **Shall** - Mandatory requirement; *should* = recommended, *may* = optional.

1.4 Functional vs. Non-Functional - How to Read This SRS

Functional requirements describe **WHAT** the system does: its features and behaviour (e.g. *"The system shall allow a user to post an item with photos"*).

Non-functional requirements describe **HOW WELL** the system does it: quality and limits (e.g. *"The search results shall appear within 2 seconds"*).

2. Overall Description

2.1 Product Perspective

The system is a new, stand-alone web application with three parts: a browser-based user interface, a server-side application (business logic), and a relational database. It replaces the current informal way of exchanging used items through scattered social media groups.

2.2 Main Product Functions

- Account management (register, login, profile).
- Item listing management (create, edit, delete, mark as completed).
- Search, browse, filter, and favourites.
- Buy / exchange requests with status tracking, and messaging.
- Ratings, reviews, and reporting of inappropriate content.
- Administration (users, listings, categories, reports).

2.3 Operating Environment

Used through a modern web browser (Chrome, Edge, Firefox, Safari) on a PC, tablet, or smartphone with an internet connection. The server runs on a standard web server with a relational database (proposed: MySQL), to be confirmed in the design phase.

3. User Roles

3.1 Guest

A visitor who is not logged in.

- Can view and search listings, register, and log in.
- Cannot send requests or see owner contact details.

3.2 Registered User

A student or community member with an account. The same account acts as *Seller/Owner* and as *Buyer/Requester* depending on the situation.

- Everything a guest can do.
- Manage own profile; post, edit, and delete own listings.
- Send and answer requests; chat; favourite items; rate users; report content.

3.3 Administrator

A staff member (admin team) who moderates the platform.

- Manage users (block / unblock) and categories.
- Approve or remove listings; review reports; view statistics.

4. User Requirements

User requirements state, in everyday language, what each type of user needs from the system. They are made detailed and testable in Sections 5-7.

UR-01	[Guest] I want to browse and search items without an account to see if the platform is useful.
UR-02	[User] I want to create an account and log in securely so my items and requests are linked to me.
UR-03	[User] I want to post an item with photos, price, and condition so others can find it.
UR-04	[User] I want to edit, delete, or mark my item as sold/exchanged so the information stays correct.
UR-05	[User] I want to search and filter by category, price, condition, and location to find items quickly.
UR-06	[User] I want to send a buy or exchange request and see its status so I know what happens next.
UR-07	[User] I want to accept or reject requests on my items so I stay in control of my exchanges.
UR-08	[User] I want to chat with the other user so we can agree on the details and a meeting place.
UR-09	[User] I want to rate users after an exchange so that trustworthy users are recognised.
UR-10	[User] I want to report fake or inappropriate listings or users so the platform stays safe.
UR-11	[User] I want my personal data and password to be protected.
UR-12	[Admin] I want to manage users, listings, categories, and reports so the platform stays clean and organised.

5. System Requirements

System requirements describe the technical environment the system needs. The functional and non-functional requirements are specified in Sections 6 and 7.

- **Client hardware** - Any PC, tablet, or smartphone able to run a modern web browser; minimum 2 GB RAM.
- **Server hardware** - Standard web server or local/cloud hosting: 2 CPU cores, 4 GB RAM, 20 GB storage (database and uploaded images).
- **Client software** - Latest two versions of Google Chrome, Microsoft Edge, Mozilla Firefox, or Safari with JavaScript enabled.
- **Server software** - Web server (e.g. Apache/Nginx), server-side language (e.g. PHP or Node.js), and a relational database (e.g. MySQL). Final choice in the design phase.
- **Network** - Internet connection (minimum 1 Mbps) for users; HTTPS for all data transfer.
- **External interfaces** - Email service (SMTP) for account verification and password reset; image upload (JPG/PNG) interface.
- **Data storage** - Relational database holding users, items, categories, requests, messages, ratings, and reports; images stored on the server file system.

6. Functional Requirements (What the System Does)

Priority: High = must be delivered; Medium = should be delivered; Low = if time allows.

6.1 Account Management

FR-01 The system shall allow a guest to register using full name, email, phone number, and password. *[High]*

FR-02 The system shall ensure that each email address is unique and shall send a verification email. *[Medium]*

FR-03 The system shall allow registered users to log in and log out. *[High]*

FR-04 The system shall allow users to reset a forgotten password via email. *[Medium]*

FR-05 The system shall allow users to view and edit their profile (name, phone, photo, location). *[Medium]*

6.2 Item Listing Management

FR-06 The system shall allow a user to create a listing with title, description, category, condition, listing type (sell / exchange / donate), price, location, and up to 5 images. *[High]*

FR-07 The system shall allow the owner to edit or delete their own listings. *[High]*

FR-08 The system shall allow the owner to mark a listing as sold, exchanged, or donated. *[High]*

FR-09 The system shall display a listing status (Available / Reserved / Completed). *[Medium]*

6.3 Search and Browse

FR-10 The system shall allow all visitors to browse listings by category. *[High]*

FR-11 The system shall provide keyword search on listing title and description. *[High]*

FR-12 The system shall allow filtering by price range, condition, listing type, and location. *[Medium]*

FR-13 The system shall allow sorting by newest, oldest, and price. *[Low]*

FR-14 The system shall show an item details page with images, description, and owner rating. *[High]*

FR-15 The system shall allow registered users to save items to a favourites list. *[Low]*

6.4 Requests and Messaging

FR-16 The system shall allow a registered user to send a buy or exchange request for an available listing. *[High]*

FR-17 The system shall allow the owner to accept or reject a request. *[High]*

FR-18 The system shall track each request status: Pending, Accepted, Rejected, Cancelled, Completed. *[High]*

FR-19 The system shall provide in-app messaging between the requester and the owner. *[Medium]*

FR-20 The system shall notify users (in-app and/or email) of new requests, replies, and status changes. *[Medium]*

6.5 Ratings and Reports

FR-21 The system shall allow both users to rate (1-5) and review each other after a completed request. *[Medium]*

FR-22 The system shall allow users to report a listing or a user with a reason. *[Medium]*

6.6 Administration

FR-23 The system shall allow the administrator to view, block, and unblock user accounts. *[High]*

FR-24 The system shall allow the administrator to approve, hide, or remove listings. *[High]*

FR-25 The system shall allow the administrator to add, edit, and delete categories. *[Medium]*

FR-26 The system shall allow the administrator to review and resolve reports. *[Medium]*

FR-27 The system shall show the administrator basic statistics (users, listings, completed exchanges). *[Low]*

7. Non-Functional Requirements (How Well the System Performs)

7.1 Performance and Capacity

NFR-01 Pages shall load within 3 seconds and search results within 2 seconds under normal load. *[High]*

NFR-02 The system shall support at least 100 concurrent users and 1,000 registered users / 5,000 listings. *[Medium]*

7.2 Security and Privacy

NFR-03 Passwords shall be stored hashed (e.g. bcrypt) and never in plain text. *[High]*

NFR-04 All traffic shall use HTTPS; sessions shall expire after 30 minutes of inactivity. *[High]*

NFR-05 Role-based access control shall prevent users from accessing admin functions or other users' data. *[High]*

NFR-06 All inputs shall be validated to prevent SQL injection and cross-site scripting (XSS); uploads limited to JPG/PNG, max 5 MB each. *[High]*

NFR-07 A user's phone number and email shall only be shown to the other party after a request is accepted. *[Medium]*

7.3 Usability

NFR-08 A new user shall be able to post an item within 5 minutes without training. *[High]*

NFR-09 The interface shall be responsive (mobile, tablet, desktop) and use consistent navigation and clear error messages. *[High]*

NFR-10 The interface shall be in English; Kurdish language support is optional for future versions. *[Low]*

7.4 Reliability and Availability

NFR-11 The system shall be available at least 99% of the time during the semester, excluding planned maintenance. *[Medium]*

NFR-12 Automatic database backup shall be done at least once per day. *[Medium]*

7.5 Maintainability, Compatibility and Scalability

NFR-13 The code shall be modular (layered / MVC), commented, and documented so that another developer can maintain it. *[Medium]*

NFR-14 The system shall work on the latest two versions of Chrome, Edge, Firefox, and Safari. *[High]*

NFR-15 The design shall allow adding new categories, features (e.g. payment), or more users without restructuring the database. *[Low]*

8. System Constraints

- **Time** - The project must be completed within the remaining weeks of the semester and follow the 9-part project schedule.
- **Team / Resources** - Only four student developers; no paid staff, tools, or services (free and open-source technologies only).
- **Budget** - No budget: free hosting or local server and free libraries.
- **Technology** - Web-based system only; no native mobile app. Must use a relational database and run in standard browsers.
- **Scope** - No online payment gateway, delivery service, or AI-based features in this version.
- **Legal / Privacy** - Personal data must be protected; prohibited or illegal items must not be allowed (enforced by admin moderation).
- **Academic** - Must follow software engineering practices and be approved by the supervisor; documents written in English.
- **Infrastructure** - Depends on available internet connectivity, which may be unstable in some locations.

9. Assumptions and Dependencies

- Users have a smartphone or computer, a web browser, and an internet connection.
- Users provide correct information and are responsible for meeting and handing over items safely themselves.
- Payment and delivery are arranged directly between users outside the system.
- An email service is available for verification and password reset emails.
- The first users will mainly be Soran University students, so the expected load is small to medium.
- At least one administrator is available to moderate content during the semester.
- The requirements may change after supervisor feedback; changes will be agreed by the group and recorded in the next version of this SRS.

10. Requirements Traceability (User Requirement to System Requirement)

- **UR-01** - FR-10, FR-11, FR-14 | NFR-01, NFR-09
- **UR-02** - FR-01 to FR-05 | NFR-03, NFR-04
- **UR-03, UR-04** - FR-06 to FR-09 | NFR-06, NFR-08
- **UR-05** - FR-10 to FR-15 | NFR-01, NFR-02
- **UR-06, UR-07** - FR-16, FR-17, FR-18, FR-20 | NFR-07, NFR-11
- **UR-08** - FR-19 | NFR-05, NFR-07
- **UR-09, UR-10** - FR-21, FR-22 | NFR-05
- **UR-11** - FR-03, FR-04 | NFR-03 to NFR-07
- **UR-12** - FR-23 to FR-27 | NFR-05, NFR-13